# MAUS User Guide

# MAUS User Guide

> **Note:** This is a placeholder ("dummy") guide scaffold. Replace the bracketed `[TODO]` sections with project-specific content before publishing.

*Methyl Assignments Using Satisfiability* — a clean-room reimplementation of the SAT-based methyl NMR assignment method of Nerli et al., *Nat. Commun.* 12:691 (2021).

---

## 1. Overview

MAUS assigns methyl NMR resonances by **subgraph isomorphism** rather than by scoring. Given a protein structure and real **peak lists** (HMQC + NOESY), it returns — for each peak — the *set* of methyls consistent with every hard constraint. It never emits a single guess unless the data force one, and it provably never excludes a valid assignment.

| Concept | In this tool |
|---|---|
| Structure graph **G** | one node per methyl carbon; geminal / short / long edges |
| Data graph **H** nodes | HMQC peaks: ($^1$H, $^{13}$C) shift + residue type |
| Data graph **H** edges | NOESY cross peaks, endpoints matched to HMQC by frequency |
| Solver | `python-sat` (Glucose3), assumption-based enumeration |
| Output | per-peak option set (1 / 2–3 / >3 / unassigned) |

---

## 2. Installation

```
python3 -m pip install python-sat
git clone https://github.com/deepnmr/maus.git
cd maus
```

Requirements:

- Python ≥ 3.8
- `python-sat` (provides the Glucose3 SAT solver)

---

# 3. Quick start

```
# 1. build peak lists from a BMRB shift file + the PDB
python make_peaklists.py examples/mbp/1ANF.pdb
        examples/mbp/bmr7114_3.str

# 2. run MAUS on the HMQC + NOESY peak lists (--truth scores against
        the key)
python maus.py examples/mbp/1ANF.pdb examples/mbp/hmqc.tsv
        examples/mbp/noesy.tsv \
    --truth examples/mbp/hmqc_true.tsv --tol-h 0.02 --tol-c 0.2 --out
        mbp_options.tsv
```

Expected summary:

```
methyls(G nodes)=192  HMQC peaks=192  NOESY cross peaks=825
NOE match: firm=508 ambiguous(dropped)=317 unmatched=0
unique(1 option)      = 130/192
ambiguous(2-3 options)= 19/192
ambiguous(>3 options) = 43/192
unassigned            = 0/192
truth in option set   = 192/192 = 100.0%
```

---

# 4. Input formats

## 4.1 PDB structure

Standard `ATOM` records. Only these residue types are parsed by default: `ALA ILE LEU MET THR VAL`. Chain `A` (or blank) is used.

## 4.2 HMQC peak list (TSV) — data-graph nodes (input)

```
label <TAB> H_ppm <TAB> C_ppm <TAB> res_type
P1      0.828        24.510       L
```

- `label` — anonymous peak id (`P1, P2, …`); leaks nothing about the answer
- `H_ppm`, `C_ppm` — methyl ($^1H$, $^{13}C$) chemical shifts
- `res_type` — one-letter code (`A I L M T V`), known from labeling

Ground truth lives in a **separate** file, never in the input:

```
label <TAB> H_ppm <TAB> C_ppm <TAB> res_type <TAB> True
P1      0.828        24.510       L              L7CD1
```

Pass it with `--truth` to score; omit it to run blind.

### 4.3 NOESY peak list (TSV) — data-graph edges

```
peak_id <TAB> H1 <TAB> C1 <TAB> H2 <TAB> C2 <TAB> mix
X1        0.828    24.51   0.712    23.10   short
```

Each row is a methyl-methyl cross peak. During the run both endpoints are matched back to HMQC peaks by frequency (within `--tol-h`/`--tol-c`); `mix` ∈ `{short, long}` tags the mixing-time distance class.

### 4.3b HMBC-HMQC peak list (TSV, optional) — geminal links

```
label <TAB> C1 <TAB> C2 <TAB> H
B1      25.39    24.56   0.340
```

One row per Leu/Val residue: a methyl proton `H` correlated to its own carbon `C1` and its geminal partner's carbon `C2`. Pass with `--hmbc`; MAUS matches endpoint A by `(H,C1)` and endpoint B by carbon `C2` (partner proton absent), then forces the pair onto a geminal structure edge. The carbon-only endpoint is degenerate (on MBP ~1/50 links resolve; the rest drop), and even a firm link couples the pair without fixing which residue — so per-peak option *counts* are unchanged on MBP. The never-exclude guarantee is preserved.

### 4.4 Generating the peak lists

`make_peaklists.py` builds all three files from a PDB and a BMRB NMR-STAR shift file (`bmrXXXX_3.str`): `hmqc.tsv` (input), `hmqc_true.tsv` (truth key), and `noesy.tsv`. HMQC ($^{1}$H,$^{13}$C) coordinates are the real BMRB methyl shifts; NOESY cross peaks are emitted for structurally close methyl pairs.

---

## 5. Command-line options

| Option | Default | Meaning |
|---|---|---|
| `--short-cut` | 6.0 | structure short-range edge cutoff (Å) |
| `--long-cut` | 10.0 | structure long-range edge cutoff (Å) |
| `--tol-h` | 0.02 | $^{1}$H NOESY/HMBC→HMQC match tolerance (ppm) |
| `--tol-c` | 0.20 | $^{13}$C NOESY/HMBC→HMQC match tolerance (ppm) |
| `--hmbc` | – | optional HMBC-HMQC geminal-link peak list |
| `--truth` | – | truth-key TSV for scoring |
| `--labeling` | `A;I;L;M;T;V` | residue types present |
| `--out` | – | write per-peak options TSV |

`make_peaklists.py` options: `--noe-short` (6.0), `--noe-long` (8.0), `--keep-k` (12 nearest NOE partners per methyl).

---

# 6. Interpreting output

The `--out` TSV columns:

| Column | Meaning |
|---|---|
| `label` | input peak identifier (anonymous) |
| `res_type` | one-letter residue type |
| `n_options` | number of valid methyls |
| `options` | comma-separated methyl labels |
| `truth` | ground-truth label (blank if no `--truth`) |
| `truth_in_set` | 1 if truth ∈ options, else 0 (blank if no `--truth`) |

- **1 option** → uniquely assigned.
- **2–3 options** → residual ambiguity, usually geminal/local symmetry.
- **>3 options** → weak local NOE network.
- **0 options** → over-constrained inputs (check cutoffs).

---

# 7. On the bundled benchmark

MAUS sees only the two peak lists. HMQC ($^{1}$H,$^{13}$C) coordinates are the real BMRB 7114 shifts, and each NOESY endpoint is resolved back to an HMQC peak **by frequency**, not by identity — so the assignment is not circular and shift degeneracy has real consequences.

- `truth in option set = 100%` is the MAUS **guarantee**: a valid assignment is provably never excluded (not a measurement — a property of the exact enumeration).
- The residual ambiguity is **real**. NOESY cross peaks whose endpoint matches more than one HMQC peak cannot be pinned to a definite methyl pair and are dropped, so shift-degenerate methyls keep several options. Tighter `--tol-h`/`--tol-c` recovers more unique calls (170/192 at ±0.01/0.1) as fewer NOEs are ambiguous — the honest resolution/degeneracy trade-off.

The NOESY cross peaks themselves are still *generated from the structure* (close methyl pairs), so this remains a controlled benchmark rather than a picked experimental NOESY. Swap in a real NOESY peak list to go fully experimental — the `maus.py` interface does not change.

---

# 8. Troubleshooting

| Symptom | Likely cause | Fix |
|---|---|---|
| `ModuleNotFoundError: pysat` | solver not installed | `pip install python-sat` |
| many `unassigned` peaks | structure cutoffs too tight | raise `--long-cut` |

| Symptom | Likely cause | Fix |
|---|---|---|
| most peaks ambiguous | too many NOEs dropped as ambiguous | tighten `--tol-h`/`--tol-c` |
| high `unmatched` NOE count | tolerance too tight vs shift precision | loosen `--tol-h`/`--tol-c` |

---

# 9. References


- Nerli, De Paula, McShan & Sgourakis. *Backbone-independent NMR resonance assignments of methyl probes in large proteins*. **Nat. Commun.** 12:691 (2021). doi:10.1038/s41467-021-20984-0

- See `COMPARISON.md` for the head-to-head against MAGIC.

---

*[TODO] Add author, license, and contact sections before release.*